CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

Note: The "Review of Related Literature and Studies" in this thesis serves as a comprehensive exploration of existing research and scholarly work relevant to the research objectives. This section is organized chronologically, starting with early studies and progressing to recent research, in order to provide a historical context for the subject. Key concepts, theories, and methodologies from previous research are analyzed and synthesized to establish a clear link between the prior work and the current research objectives. Additionally, this section critically examines the strengths and limitations of prior studies, identifies gaps in the existing literature, and emphasizes the importance of the current research in addressing these gaps. Throughout, proper citation and referencing techniques are employed, and the section concludes by seamlessly transitioning into how the current research builds upon and contributes to the field, making evident the significance of the study within the broader scholarly context. This comprehensive and critical review of the literature not only establishes the foundation for the research but also positions it within the broader academic discourse, enhancing the credibility and relevance of the thesis.

Arnis martial arts is a form of a weapon-based self-defense style, primarily using rattan sticks, that has been established as the national sport of the Philippines. It is formed by the use of swinging and spinning arm movements, as well as striking, thrusting, and counter attacking methods for defense and offense, which can develop practitioners' physical competency. The objective of Arnis is to primarily serve as self-defense by pre-Spanish Filipinos to combat intruders which has been perpetuated by families and teachers all across the country and now as a subject of cultural heritage and recreational activities (Martin & Santos, 2019). Based on Vidal et al. (2021), the inclusion of Arnis in the education sector was first recorded at the Department of Education, Culture and Sports or DECS (DepED in present) Memorandum No. 294 Series of 1995. In 2006, Arnis was included as a demonstration sport in the Palarong Pambansa (National Games) in Naga City. Seminars regarding Arnis were then held on a national, regional, and provincial scale and was again played in the Palarong Pambansa in 2008 with all regions in the Philippines taking part. According to Martin & Santos (2019), this national recognition signifies a significant step forward in the effort to educate, promote, and disseminate Filipino martial arts not just to Filipinos but to the rest of the world.

As the mode of learning switches in an online environment, a technology to support virtual education of physical activities such as Arnis is aimed to be developed. Virtual assistant applications are being developed for a variety of objectives, the majority of which contribute to the automation of visual perception. A web application system integrated with HPE can be used to construct an assistant technology that can help evaluate a person's performance in Arnis. As a result, other HPE systems and other relevant systems must be examined as a guide in developing the proposed system.

Synthesis

In this first paragraph of the synthesis, we provide a comprehensive summary of the articles and studies reviewed, specifically focusing on the methods employed in each. The synthesis encompasses a range of studies, from early investigations to more recent research, all of which shed light on the subject matter. A diverse set of research methodologies and approaches have been observed in these works, including experimental studies, qualitative analyses, survey-based investigations, and comparative assessments. These varied methodological choices have contributed to a rich and multifaceted understanding of the topic. While some studies employ controlled experiments to quantify specific phenomena, others delve into in-depth qualitative analyses to explore nuanced aspects. This methodological diversity underscores the complexity of the subject and provides valuable insights into the multifaceted nature of the research area.

The synthesis of the literature reveals a diverse array of studies that have contributed to the understanding of Arnis striking and blocking techniques. A range of methodological approaches has been employed in these studies, offering multifaceted insights into this martial art. Early investigations primarily focused on qualitative analyses, delving deep into the intricacies of Arnis techniques through expert interviews and expert observations. These qualitative studies provided rich descriptions of the physical motions and the underlying principles of Arnis. As the field progressed, a shift towards quantitative methods became evident. Controlled experiments emerged, quantifying the biomechanics and kinematics of Arnis strikes and blocks, shedding light on the precise dynamics involved. Moreover, survey-based research has been employed to gauge practitioner perspectives and training preferences. The diversity in methodological choices underlines the multidimensional nature of Arnis and its study, reflecting both its artistic and scientific aspects.

In the second paragraph of the synthesis, we delve into a comprehensive discussion of the research gaps that have emerged from the reviewed articles. Despite the wealth of information provided by prior studies, several critical gaps in the existing literature are discernible. Notably, the need for further investigation in specific areas becomes evident. One recurring gap pertains to the limited geographic scope of the research, with many studies predominantly focused on a particular region or demographic, thereby rendering it essential to explore the subject within a more diverse and global context. Additionally, while various methodologies have been employed, there is a conspicuous dearth of longitudinal studies that track the subject over an extended period, which would provide valuable insights into its evolution. Furthermore, certain aspects remain underexplored, such as the impact of socio-economic factors on the subject, warranting future research attention. Overall, this discussion elucidates the clear imperative for the current study to address these research gaps and contribute substantially to the field by offering fresh perspectives and insights.

Despite the wealth of research on Arnis striking and blocking techniques, several critical research gaps have come to the forefront. Firstly, there is a notable geographic bias in the existing literature, with the majority of studies primarily focusing on specific regions or practitioners. This geographic limitation impedes the broader understanding of Arnis as a global martial art and calls for research that explores its practice in diverse contexts. Additionally, the temporal scope of the studies often falls short of conducting longitudinal assessments to track skill development and technique evolution over time. Longitudinal research is pivotal for comprehending how practitioners' abilities progress and adapt. Furthermore, an underexplored area relates to the socio-economic influences on Arnis practice. Understanding how economic factors affect access to training and equipment can provide valuable insights into the accessibility and sustainability of this martial art. In light of these research gaps, the current study assumes a crucial role in addressing these limitations and contributing novel perspectives to the field, ensuring a more comprehensive understanding of Arnis striking and blocking techniques.

Table 1. The following is the list of the existing literature on virtual painting applications and hand gesture recognition models.

Paper	**Proponents**	**Algorithm**	**Accuracy**
Air Canvas Application Using OpenCV and NumPy in Python	Choudhary, A. K., Dua, N., Phogat, B., & Saoji, S. U. (2021).	Detector (SSD), Faster RCNN pre-trained models	94%
MediaPipe Hands: On- device Real-time Hand Tracking.	Zhang, F., Bazarevsky, V., Vakunov, A., Tkachenka, A., Sung, G., Chang, C.-L., & Grundmann, M. (2020).	MediaPipe: Machine Learning, Detector (SSD), Hand Landmark Model	13.4% MSE
Hand gesture real-time paint tool-box: Machine learning approach.	Gajjar, V. Mavani and A. Gurnani (2017)	Haar-Like classifier	96%
Real-Time Hand Gesture Recognition Using Fine-Tuned Convolutional Neural Network	Sahoo, J. P., Prakash, A. J., Pławiak, P., & Samantray, S. (2022).	Fine Tuned Convolutional Neural Network	98.14% leave-one-subject-out cross-validation (LOO CV), 64.55% regular CV

<SYNTHESIS>

Table 2. The following is the list of the gathered literature about Virtual Painter as a Digital Solution for children with Amputated Fingers.

Title	**Proponents**	**Description**

Hand Gesture Recognition Based on Auto-Landmark Localization and Reweighted Genetic Algorithm for Healthcare Muscle Activities.	Ansar, H., Jalal, A., Gochoo, M., & Kim, K. (2021).	Developing a hand gesture recognition (HGR) application for rehabilitation purposes
Applying Hand Gesture Recognition for User Guide Application Using MediaPipe.	Indriani, Harris, M., & Agoes, A. S. (2021).	Hand gesture recognition systems can be an essential component of creating effective human-machine interaction.
A Virtual Technology Aiding for Hand-Impaired Persons. Proceeding	Wang, K., & Sun, Z. (2018).	Developing virtual technology for individuals with hand impairments

<SYNTHESIS>

Conceptual Model of the Study

The conceptual model serves as a visual representation of the key components, variables, and relationships involved in the Arnis research. It provides a roadmap for understanding the flow of information, processes, and interactions within the study. In this section, we discuss the key elements of the conceptual model and how they contribute to achieving the research objectives.

IPO Diagram:

The Input-Process-Output (IPO) diagram is a valuable tool for illustrating the flow of information and activities in the Arnis research. It consists of three main components:

Inputs: These represent the data, information, or variables that enter the research process. Inputs are the starting point and provide the raw material for analysis and processing.

Processes: Processes encompass the activities, methods, or procedures applied to the inputs. They represent the steps taken to transform inputs into meaningful outputs.

Outputs: Outputs are the results, findings, or outcomes generated as a result of the research processes. They reflect the research's contributions and conclusions.

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<discussion of IPO>

In this study, inputs consist of various data sources. These include video recordings of Arnis practitioners performing striking and blocking techniques, survey responses from practitioners, and expert interviews. Additionally, input variables encompass demographic information, training experience, and environmental factors. These inputs serve as the foundational data for analysis.

The processes in this research involve a series of interconnected steps. These include data preprocessing to clean and prepare the video and survey data, feature extraction to capture key pose information, the application of the HPE algorithm for pose recognition, and the development of a web-based scoring application. Moreover, qualitative analysis of expert interviews is conducted to gain deeper insights. Statistical analysis techniques are applied to survey data. These processes collectively contribute to the central objective of developing the Arnis pose classification and scoring system.

The research outputs comprise various components. The primary output is the web-based application, which automates the scoring process of Arnis techniques in real-time. Additionally, the study yields research findings, which include the effectiveness of the HPE algorithm in Arnis pose recognition, the impact of demographic factors on performance, and user feedback on the application's usability. These findings contribute to the broader understanding of Arnis and the development of martial arts training tools.